

PLANTAIN OUTGROWERS
PORTAL

FIELD OFFICER EDITION · V1.0

Plantain Procurement & Grading

A field guide for buying and grading
True Horn & False Horn plantain

Bitverse · Plantain Outgrowers Portal

Bitverse Agro · Nairobi · June 2026

For internal Field Officer use

Cover photo: Wikimedia Commons

FOREWORD

Buying plantain is not just weighing.

This guide exists because the difference between a fair price and a cheated one — between a good bunch and a rejectable one — lives in the field, at the moment of weighing, in the eye and the hand of the Field Officer.



A plantain market in East Africa — the context where procurement decisions land.

The Plantain Outgrowers Portal (POP) buys cooking plantain — False Horn and True Horn — from smallholder outgrowers across our network, then resells into urban wholesale markets. Every kilogram we procure passes through a Field Officer who decides its grade, its price, and whether it moves at all.

That decision has three downstream effects:

WHAT A FIELD OFFICER'S CALL DECIDES

- **The outgrower's income** — a Grade A pays KES 20/kg more than a Grade C. The same bunch, two readings.
- **Our market's reputation** — bad grading at intake becomes a complaint at the wholesale floor.
- **The Portal's integrity** — every rejection must be defensible. Every acceptance must be defensible.

So this is not a manual. It is a record of how POP expects a Field Officer to read a bunch, talk to an outgrower, weigh, record, and decide. Read it once before your first procurement. Read it again before your first dispute.

"Reject the bunch, not the outgrower. Document every rejection. Defend every acceptance."

How to use this guide

Before each visit. Read the section that matches the task — pricing band, shrinkage rules, defect catalog.

During the visit. Keep the tear-off quick card on the last page in your pocket.

After the visit. Use the rejection language in Section 10 if any bunch is refused.

When in doubt. Photo + measure + record. Disputes resolve on data, not memory.

This is version 1.0, June 2026. As POP grows — new varieties, new markets, new procurement zones — this guide will grow with it. Send corrections and field notes to your supervisor. The next edition is built from this one.



01

SECTION ONE

Role & Mindset

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Role & Mindset

The Field Officer is not a buyer, not a salesperson, and not a quality inspector who happens to also pay. The role is a single thing: **the trusted reader of the bunch, in the outgrower's field, on POP's behalf.**

The three overriding rules

1. THE PRICE IS SET BEFORE YOU ARRIVE.

The day's reference price was pulled from the app at 6 am. Your job is to apply it within the $\pm 5\%$ band — not to negotiate the price itself. If the outgrower wants a different price, the answer is "the daily price is what I pay; I can discuss the grade, which moves the price."

2. THE GRADE IS WHAT THEY BROUGHT, NOT WHAT YOU NEED.

Do not adjust a grade to meet a quota. Do not upgrade a B to an A to be polite. If you cannot defend the grade on a photo and a measurement, the grade is wrong.

3. THE RELATIONSHIP OUTLIVES THE BUNCH.

This outgrower will be selling to POP next month, next season, next year. A bad day at the scale costs more than one bunch's worth of savings. Be firm on grading. Be fair on price. Be polite, always.

First 30 seconds

When you arrive, you set the tone for the entire visit. Three things happen in the first 30 seconds:

1

GREET, BY NAME IF YOU CAN.

"Jambo, Mama Wanjiku. I'm here for the bunch you signaled." Using the outgrower's name — or asking for it — is the cheapest relationship investment you'll make all week.

2

STATE THE DAY'S REFERENCE PRICE.

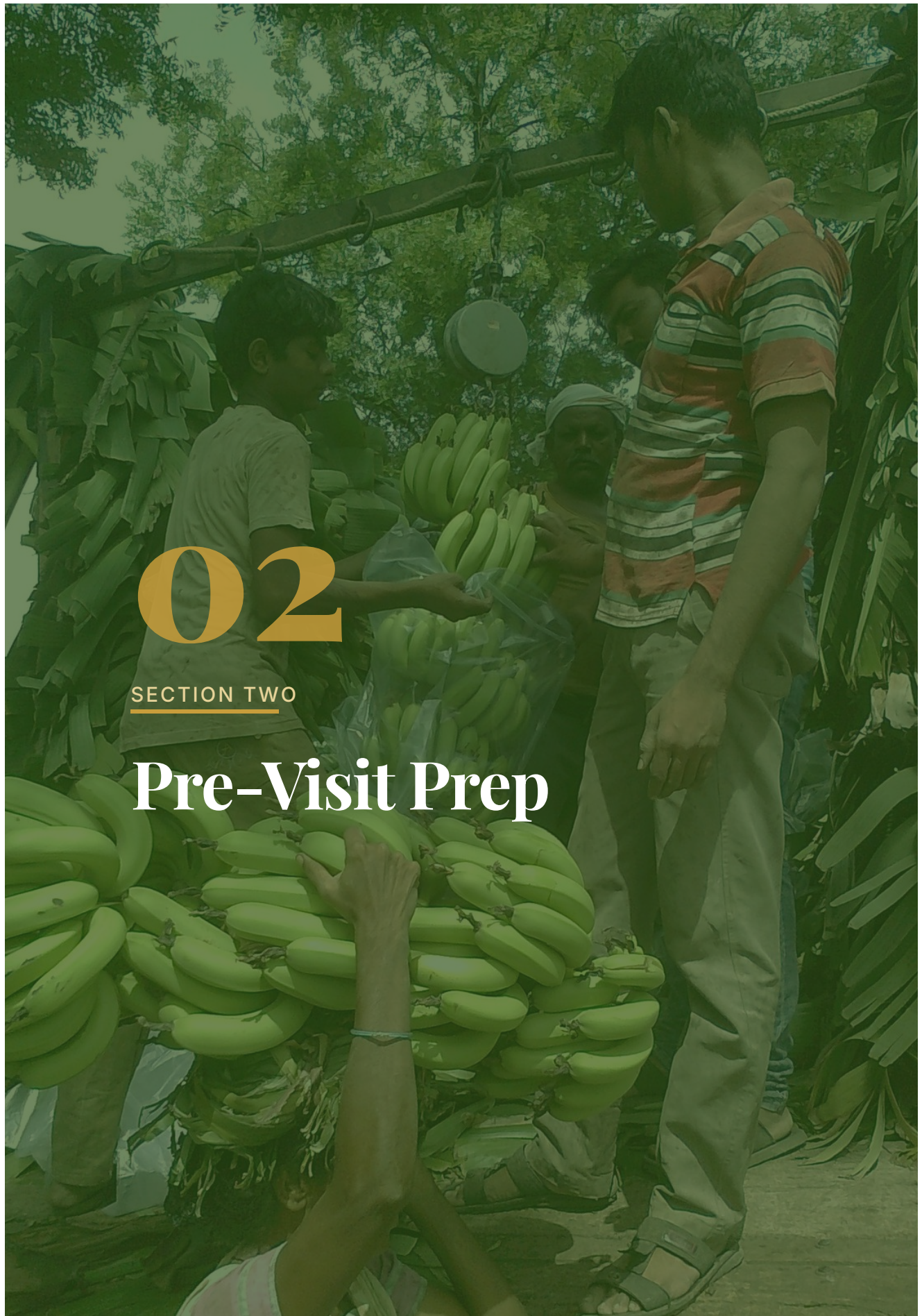
"Today's price for Grade A is KES 32/kg. I'll weigh, grade, and pay against that." Outgrowers who know the price in advance trust the transaction. Outgrowers who don't, suspect the transaction.

3

ASK PERMISSION TO WALK THE FIELD.

"I'd like to see the bunches before we start." This is professional, not invasive. It also lets you spot whether the bunches in the signal are the best ones or the worst ones.

"You are POP at the farm gate. Everything about how POP is perceived — fairness, competence, care — is read off you in the first 30 seconds."



02

SECTION TWO

Pre-Visit Prep

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Pre-Visit Preparation



Every visit starts the night before. A Field Officer who arrives unprepared pays for it in slower decisions and weaker records.

The night before

- **Pull today's reference price** from the POP app. Note it. Screenshot it. The app is the source of truth.
- **Review this morning's supply signals** — who signaled, what they signaled, what grade they self-reported.
- **Check the route** — order your visits by zone to minimize backtracking. Bunches bruise with every kilometer.
- **Charge your phone** to $\geq 80\%$. You'll use it for photos, scale pairing, and the procurement form.

The morning of

THE 8-TOOL FIELD KIT

Show up with these. Anything missing is a procurement that doesn't happen.

1. **Calibrated digital scale** (50 kg min, 100 g precision) — battery fresh, paired to phone
2. **Sharp, clean knife** — for cutting the bunch stem
3. **Latex cup / collection pan** — to drain the sap before weighing (saves 2–4% mass)
4. **Tare weight** — known hook + sling weight, written on the sling
5. **Sample bag** — to carry rejected bunches' sample fingers for documentation
6. **Recording device** — phone with POP app + offline form backup
7. **Permanent marker + tag ties** — to mark accepted bunches by grade
8. **This guide, in your pocket** — especially the tear-off card on the last page

Before you leave the depot

CHECK	WHY
Scale zeros at 0.000 kg with no load	Prevents systematic 200 g over-read
Reference weight (e.g., 10 kg) reads within ± 50 g	Calibration drift happens in transit
Phone has signal at the first two farms	If not, fall back to offline form
Cash float for spot payments ($\leq$ KES 5,000)	M-Pesa is preferred; cash is the fallback
Vehicle fuel $\geq$ half tank	Stranded FO = stranded bunches

SECTION THREE

The Visit Workflow

A procurement visit has six steps. Most take under 90 minutes total. The order matters — re-ordering them is how disputes happen.

1**ARRIVE, GREET, STATE THE PRICE.**

First 30 seconds — see Section 1. Name. Price. Permission to walk the field.

2**WALK THE FIELD AND COUNT BUNCHES.**

Confirm what the outgrower signaled. Note any that look diseased, over-mature, or damaged. These are flagged for closer inspection later.

3**SAMPLE-INSPECT AT LEAST 1 BUNCH IN 5.**

Don't grade from outside the bunch. For every 5 bunches, cut one open and check finger condition. Disease hides inside.

4**GRADE EACH ACCEPTED BUNCH, MARK WITH TAG.**

Use the threshold table in Section 6. Tie a colored tag: green = A, yellow = B, red = C, white = reject. The tag stays with the bunch to the depot.

5**WEIGH, RECORD, PHOTOGRAPH.**

Net weight = gross – tare. Record in the app. Photograph the bunch on the scale, showing the weight reading.

6**PAY, SIGN, CONFIRM, LOAD.**

M-Pesa first, cash fallback. Outgrower signs (or thumbprints) the form. Confirm pickup time. Load carefully — no drops, no crush.

ON LATEX DRAINAGE

A freshly cut bunch bleeds 2–4% of its mass as latex within the first 30 minutes. Always weigh after drainage, never before. If you weigh immediately, you will pay for water that won't be in the bunch at the depot — and shrinkage will trigger on a perfectly fine load.

03

SECTION FOUR

Varieties

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Varieties — True Horn & False Horn

POP procures only two varieties of cooking plantain: **False Horn** and **True Horn**. Both are Musa AAB genome, both cook green, both are staples across East and West Africa. They are not interchangeable — the bunch shape, finger count, and market price differ. Mixing them silently is how grading errors and outgrower disputes both start.

THE SINGLE RULE

Grade within a variety. Never compare a False Horn finger to a True Horn finger on the same ruler. Each variety has its own threshold table — see Section 6.

Side-by-side

Banana varieties comparison

A mixed display of cooking banana varieties — False Horn on the left with its incomplete hand, True Horn on the right showing the full complement of hands.

African plantain varieties

False Horn varieties commonly grown in East Africa — note the larger individual fingers and the incomplete (horn-like) final hand.

Identification cheat sheet

TRAIT	FALSE HORN (FRENCH)	TRUE HORN
Hands per bunch	5–8 hands + a final incomplete "horn" of 1–4 fingers	5–8 hands + a true horn of 1–2 fingers (smaller than False Horn's)
Finger size	Larger, fatter (15–25 cm, girthier)	Smaller, longer (10–18 cm, slimmer)
Bunch weight	15–30 kg typical	8–15 kg typical
Market premium	Standard reference	+5–10% (preferred for frying, restaurant trade)
Look for	Big "fist" of fingers with one trailing horn	Smaller, more uniform, slimmer horn

COMMON MISIDENTIFICATION

A young False Horn, with the horn not yet emerged, can look like a True Horn. **Always check the crown of the bunch** — if there is a small bud visible where a final hand is forming, the variety is False Horn. If the bud is a single finger or absent, it is True Horn.

SECTION FIVE

Maturity — The 9-Stage Scale

Ripe and green plantain side by side

The two ends of the maturity scale — fully green (Stage 1) on the left, fully black-ripe (Stage 8) on the right. POP procures Stages 1-2 only.

Maturity is graded on a 9-stage scale defined by peel color, finger firmness, and starch content. POP procures at **Stages 1-2 (green mature)** only. Stage 3+ is for ripening rooms and direct retail — not procurement.

STAGE	PEEL COLOR	FINGER FEEL	PROCUREMENT?
1 — Green mature	Dark green, no yellow	Hard, no give	🟢 Buy
2 — Green turning	Green with first yellow tinge at ridges	Hard, slight give at stem end	🟢 Buy (premium)
3 — Yellowing	More yellow than green	Softening, easy to peel	⚠️ Reject (ripening, not procurement)
4 — Yellow with black spots	Yellow dominant, brown flecks	Soft, finger feels "ripe"	🔴 Reject
5+ — Black ripe → overripe	Black or split peel	Yielding, smell sweet	🔴 Reject (retail / cooking loss)

WHY STAGE 1-2 ONLY

A plantain harvested at Stage 3 will ripen to black in 3–5 days at ambient. By the time it reaches the wholesale floor it is unsellable — and shrinkage flags a load that looked fine at intake. Stage 1–2 fruit gives us 7–10 days of usable life at room temperature.

"Maturity is graded by feel, not by calendar. A bunch in warm lowland ripens faster than the same variety at altitude. Trust the finger, not the date."

04

SECTION SIX

Grading A · B · C · Reject

SECTION SIX

The Grading Standard

Single plantain bunch

A Grade A bunch — full, even, clean, fingers uniformly green mature. This is the benchmark the threshold table measures against.

Proposed thresholds (June 2026)

GRADE	FALSE HORN BUNCH	TRUE HORN BUNCH	FINGER CONDITION	PRICE VS. REFERENCE
A — Premium	≥ 20 kg	≥ 12 kg	Finger ≥ 20 cm (FH) / ≥ 25 cm (TH); clean, no defects	Reference (100%)
B — Standard	12–20 kg	8–12 kg	Finger 15–20 cm (FH) / 20–25 cm (TH); minor cosmetic only	–5%
C — Lower / process	< 12 kg	< 8 kg	Finger < 15 cm (FH) / < 20 cm (TH); visible defects OK, no disease	–10% to –15%
Reject	Any weight	Any weight	Disease (BSV, sigatoka, weevil), stage 3+, severe damage, rot	No payment

TRUE HORN PREMIUM

Because True Horn commands a 5–10% premium in the restaurant and frying trade, add this on top of the grade band. Example: True Horn Grade A at +5% premium, KES 32 base → KES 33.60/kg.

How to assign a grade

For each bunch, ask four questions in order. The first "no" stops the grade from going higher.

1

REJECT TEST — IS THERE A REJECT CONDITION?

Disease? Pest? Stage 3+? Rot? Severe mechanical damage? If yes → **Reject**. Stop here.

2

WEIGHT TEST — DOES IT MEET GRADE A OR B WEIGHT?

Use the variety-specific threshold. If the bunch is below B weight, it is Grade C.

3

FINGER TEST — DOES IT MEET THE FINGER LENGTH FOR THE GRADE?

Sample 3 fingers from the top, middle, and bottom hand. Measure along the outer curve. If any finger falls short, downgrade one grade.

4

CLEANLINESS TEST — WHAT DEFECTS ARE VISIBLE?

Use the defect catalog in Section 7. Count and assess. Cosmetic alone stays A. Functional defects (latex, scarring affecting the peel) drop to B. Multiple defects → C.

VOLUME BONUS

A single load meeting or exceeding **200 kg of Grade A equivalent** (where A=1.0, B=0.95, C=0.85, reject=0) triggers a **+5% bonus** on the load total. Apply it on the form, not the per-bunch calculation.

05

SECTION SEVEN

Defect Catalog

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Defect Catalog & Responses

Banana leaf disease

Banana diseases travel from leaf to fruit. A field showing leaf symptoms will produce diseased bunches — sample-inspect at the source, not at the depot.

The 8 defects to grade on

1. BLACK SIGATOKA

Symptoms: Black streaking on peel, premature ripening in patches, leaf streaks visible in field.

Action: Reject the bunch. Photograph the leaf and the finger. Reject all fruit from that mat for the visit.

2. BANANA WEEVIL DAMAGE

Symptoms: Tunnels in the corm, weakened stem, finger separation at the crown.

Action: Reject. Mark the mat for the supervisor's follow-up — weevil spreads.

3. LATEX STAINING

Symptoms: Black-brown stains on fingers from sap oxidizing on the peel.

Action: Cosmetic only if surface. Reject if staining covers >30% of any finger or penetrates the peel.

4. MECHANICAL DAMAGE

Symptoms: Bruising, cuts, finger separation, broken crown.

Action: B if minor. C if multiple fingers affected. Reject if the crown is broken — it won't store.

5. THRIPS SCARRING

Symptoms: Raised silvery-brown corky patches on the peel between fingers.

Action: B if isolated. C if pervasive. Not a reject — purely cosmetic.

6. SUNBURN / SCALD

Symptoms: Yellow or pale patch on the sunward side of the bunch.

Action: B. Cosmetic only. Does not affect storage or cooking quality.

7. CROWN ROT / ANTHRACNOSE

Symptoms: Black soft rot at the crown, fingers detaching, dark lesions.

Action: Reject. This is the most common cause of whole-load loss at the depot.

8. BUNCHY TOP / VIRAL

Symptoms: Stunted, narrow leaves in field; small, misshapen fingers.

Action: Reject. Report the mat to the supervisor — viral diseases are notifiable.

"A defect is not a judgment on the outgrower. It is a description of the bunch. Use neutral language: 'the peel shows latex staining,' not 'your bunch is stained.'"

SECTION EIGHT

Pricing & Negotiation

Pricing is set in the app, refreshed daily. Your role is to apply it. The only lever a Field Officer controls is the grade.

The $\pm 5\%$ band

HOW THE BAND WORKS

Today's reference price for False Horn Grade A is, say, **KES 32/kg**. You may pay anywhere from **KES 30.40 to KES 33.60** — that is the field band. Above the band requires supervisor approval. Below the band is treated as a special concession and flagged in the form.

SCENARIO	ACTION
Outgrower offers load well above Grade A, all clean	Pay the band top (+5%), note as "premium load" on form
Outgrower offers load mixed A/B/C	Weigh and grade each bunch, no negotiation on the per-kg rate
Outgrower demands above-band rate	Refer to supervisor in-app. Do not exceed band in the field.
Outgrower offers below-band rate to clear stock	Acceptable up to 5% below. Document the reason (e.g., "clearing pre-mature stock").

Worked example

Outgrower: Mama Wanjiku, Kapsabet zone. **Signal:** 4 bunches, ~80 kg.

Reference (today): False Horn Grade A = KES 32/kg. True Horn premium +5%.

BUNCH	VARIETY	WEIGHT	GRADE	RATE	VALUE
1	False Horn	22 kg	A	KES 32.00	KES 704.00
2	False Horn	18 kg	B	KES 30.40 (−5%)	KES 547.20
3	True Horn	11 kg	A	KES 33.60 (+5%)	KES 369.60
4	False Horn	9 kg	C	KES 28.80 (−10%)	KES 259.20
Total		60 kg			KES 1,880.00

60 kg **does not** meet the 200 kg Grade A-equivalent threshold for the volume bonus. The 5% bonus does not apply. If the load had been 200 kg at 95% Grade A equivalent, the bonus would be $5\% \times \text{KES } 6,400 = \text{KES } 320$ extra.

SECTION NINE

Weighing, Shrinkage & Records

Net weight = gross – tare

Always subtract. The sling + hook + any packaging you used to suspend the bunch. Write the tare on the sling in marker so it stays with the scale.

Shrinkage — the $\pm 5\%$ flag

WHAT SHRINKAGE MEASURES

Shrinkage is the difference between the weight you recorded at the farm and the weight at the depot, expressed as a percentage. Normal range: 0–3% (latex loss, handling). Above 5% triggers an investigation.

SHRINKAGE	LIKELY CAUSE	ACTION
0–2%	Normal latex drainage	None — record the actual weight
2–5%	Some weight loss in transit	Note the cause on the form (heat, time, handling)
5–8%	Excessive drainage or partial ripeness	Flag for supervisor review. Was the grade correct?
> 8%	Likely Stage 3+ at intake, or rough handling	Investigate. May require a field re-training for the FO.

The procurement form — what to capture

- **Outgrower ID** — from the POP registry, not from memory
- **Date and time of weighing** — auto-stamped by the app
- **GPS pin** — auto-stamped by the app; flags visits outside registered farms
- **Bunch count** and per-bunch **variety, weight, grade**
- **Reference price** and per-bunch **rate applied**
- **Total payable** and **payment method** (M-Pesa / cash)
- **Outgrower signature or thumbprint**
- **Photo evidence** — one per rejected bunch + one bulk shot of the load
- **Notes** — outgrower comments, weather, any deviation from the band

SECTION TEN

Rejections, Disputes & Escalation

Rejections will happen. The way they are handled determines whether the outgrower comes back.

How to reject a bunch

THE 4-STEP REJECTION SCRIPT

- 1. State the defect factually.** "This bunch shows crown rot at the base — the fingers are starting to separate."
- 2. Show, don't just tell.** Point to the specific defect. Offer to cut a finger open to demonstrate.
- 3. Reference the standard.** "Per POP grading, crown rot at this stage is a reject — it would lose 30% of the load in transit."
- 4. Offer the path forward.** "The other three bunches are good A and B. We can proceed with those, and I can mark this mat for the supervisor to advise on next season."

Escalation ladder

LEVEL	TRIGGER	WHO TO CALL	WHEN
1	Outgrower disputes a single grade	Field Officer, in the field	On the spot, with photo evidence
2	Outgrower refuses all load	Field Supervisor (call)	Within 1 hour of leaving farm
3	Outgrower alleges FO misconduct	Field Supervisor + HR note	Same day, written report
4	Suspected disease / pest in field	Field Supervisor + Agronomist	Same day, photo + GPS pin
5	Safety threat / refusal to leave	Withdraw, call operations	Immediately

"A rejected bunch is a data point. A rejected outgrower is a system failure. When in doubt, escalate. Never burn a relationship to win a single procurement."

All escalations logged in the app with photos and notes. The supervisor reviews within 24 hours and either confirms the FO's call, adjusts the grade, or arranges a follow-up visit. Outgrowers always receive a written response within 48 hours — verbal-only resolutions are not acceptable.

Plantain Grading at a Glance

Cut along the dashed line. Keep in pocket during visits.

GRADE THRESHOLDS

GRADE	FH KG	TH KG
A	≥ 20	≥ 12
B	12–20	8–12
C	< 12	< 8
Reject	any	any

PRICE ADJUSTMENTS

- Grade A: reference (100%)
- Grade B: −5%
- Grade C: −10% to −15%
- True Horn: +5% to +10%
- Volume bonus (≥ 200 kg A-eq): +5%
- Field band: ±5% of reference

MATURITY — BUY STAGES

- 🟢 Stage 1 (green mature)
- 🟡 Stage 2 (green turning)
- 🟠 Stage 3+ (yellowing onward)

REJECTION TRIGGERS

- Black sigatoka on fruit or leaf
- Banana weevil (corm tunnels)
- Crown rot / Anthracnose
- Bunchy top / viral
- Stage 3+ maturity
- Severe mechanical damage

FIELD PROCEDURE

1. Greet + state today's price
2. Walk the field, count bunches
3. Sample-inspect 1 in 5
4. Grade, tag (A=green, B=yellow, C=red)
5. Drain latex, weigh, photograph
6. Record, pay, sign, load

SHRINKAGE

- 0–2% — normal
- 2–5% — note cause
- > 5% — flag for review

ESCALATION

- Grade dispute → resolve in-field with photo
- Load refused → call Field Supervisor within 1 hr
- Disease suspected → Agronomist same day

